# Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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## Research Interests

My research lies at the intersection of software engineering and human–computer interaction. I study how developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases: how they communicate with AI, how they verify and integrate AI-generated code, and whether faster, more correct work also means deeper code comprehension. Through controlled experiments, screen-recorded behavioral analysis, prompt analysis, and interviews, I derive design implications for future AI coding agents and developer tools.

## Education

**Oregon State University** — Corvallis, OR, United States
*Ph.D. in Computer Science, School of EECS* — *Sep 2023 – Jun 2027 (expected)*
**Advisor**: Dr. Christopher Hundhausen at the SEAL Lab
**Research Areas**: Empirical Software Engineering, Human–Computer Interaction, Human–AI Collaboration, Code Comprehension

**Arizona State University** — Tempe, AZ, United States
*M.S. in Software Engineering* — *Aug 2021 – May 2023*
**Advisor**: Dr. Robert LiKamWa
**Thesis**: Networked System for Volumetric Athletic Coaching in Augmented Reality

**Guangxi University of Science and Technology** — Liuzhou, China
*B.E. in Software Engineering* — *Sep 2016 – Jun 2020*

**Southern Cross University** — Gold Coast, Australia
*B.E. in Information Technology* — *Sep 2017 – Jun 2020*

## Publications (Google Scholar Page)

### Peer-Reviewed Journal Articles

**[J2]** Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, Sanjay Chandrasekaran, **Yunhan Qiao**, and Summit Haque (2026). Using SlackBot-prompted reflections to promote contribution equality in team software development projects. *ACM Transactions on Computing Education*. https://doi.org/10.1145/3857221

**[J1]** **Yunhan Qiao**, Md Istiak Hossain Shihab, and Christopher Hundhausen (2026). A systematic literature review of the use of GenAI assistants for code comprehension: Implications for computing education research and practice. *ACM Transactions on Computing Education, 26*(2), Article 24, 1–33. https://doi.org/10.1145/3785366
*Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.*

### Peer-Reviewed Conference Proceedings

**[C2]** Manish Motwani, Aakash Kulkarni, **Yunhan Qiao**, Matthew Davis, and Ziyan Chen (2025). LLM-guided differential fuzzing for detecting platform-specific bugs in scientific applications. *2025 IEEE Conference on AI x Software Engineering (AIxSE)*, 59–66. https://doi.org/10.1109/AIxSE64906.2025.00015

**[C1]** Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, **Yunhan Qiao**, and Brian Wise Mulanda (2025). The effects of GitHub Copilot on computing students' programming effectiveness, efficiency, and processes in brownfield coding tasks. *Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER '25), 1*, 407–420. https://doi.org/10.1145/3702652.3744219

### Preprints

**[P4]** Olivia Schutz, Rudrajit Choudhuri, **Yunhan Qiao**, Fatima A. Moussaoui, Julia Russo, Zoe Steine-Hanson, and Anita Sarma (2026). The vibe paradox: What EEG reveals about your brain when vibe coding. https://arxiv.org/abs/2611.02087

**[P3]** Summit Haque, Md Istiak Hossain Shihab, **Yunhan Qiao**, and Christopher Hundhausen (2026). How do computing students solve problems with AI? A cross-national study.

**[P2]** **Yunhan Qiao**, Summit Haque, and Christopher Hundhausen (2026). Beyond the prompt: Linking what developers ask, do, and understand with coding agents. https://arxiv.org/abs/2609.33712

**[P1]** **Yunhan Qiao**, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen (2025). Faster but not wiser: GitHub Copilot decouples programming performance from code comprehension in brownfield tasks. https://arxiv.org/abs/2511.02922

## Teaching Experience

**Graduate Teaching Assistant, School of EECS** — Oregon State University
CS 362: Software Engineering II
CS 461: Senior Software Engineering Project

CS 467: Online Capstone Project
CS 492: Mobile Software Development

## Professional Experience

**Software Engineering Intern, Mocha Software** Nanjing, China
*Software development* *Aug 2020 – Nov 2020*
Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

## Awards & Scholarships

Oregon State University EECS Graduate Scholarship *2023 – current*
Guangxi University of Science and Technology Merit Award *2018*

## Academic Service

**Program Committee**
ACM Technical Symposium on Computer Science Education (SIGCSE TS), 2027
IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2027

**Reviewer**
ACM Transactions on Computing Education (TOCE), 2027

## Skills & Coursework

**Research Methods**: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
**Programming**: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
**Frameworks & Tools**: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
**Coursework**: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

## Languages

English (Fluent), Mandarin Chinese (Native)